This section begins with a consideration of the biochemical pathways of photosynthesis and its importance as a synthetic process upon which most food chains and food webs are based.

Learners gain an appreciation of the role of microorganisms in recycling materials within the environment and maintaining balance within ecosystems. Microorganisms are also used as natural biosensors by environmental biologists to detect and measure the presence of chemical compounds in the environment.

In this section, the need to conserve environmental resources in a sustainable fashion is considered, whilst appreciating the potential conflict arising from the needs of an increasing human population. Learners also consider the impacts of human activities on the natural environment and biodiversity.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

4.3.1 Photosynthesis, food production and management of the environment

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** the ultrastructure of the chloroplast	To include the location of the components for the light-dependent reaction in the thylakoid membranes (light harvesting complexes, photosystems, electron transport chain and ATP synthase) and the location of the enzymes for the light-independent reactions in the stroma.
	(ii) practical investigation into the separation of pigments by paper chromatography	No details of the structure of carotenoids or accessory pigments are required. *M1.1, M1.2, M1.3, M1.6, M1.7, M1.8* **PAG6** HSW4, HSW5, HSW6
(b)	the process of the light-dependent stage of photosynthesis	To include the transfer of light energy to chemical energy in the form of ATP and reduced NADP (details of cyclic and non-cyclic photophosphorylation are not required). HSW8

(c)	the production of complex organic molecules in the light-independent stage of photosynthesis (Calvin cycle)	To include the use of the products of the light-dependent reactions (ATP and reduced NADP) within the light-independent reactions **AND** the role of carbon dioxide, ribulose bisphosphate carboxylase (RuBisCO), ribulose bisphosphate (RuBP), glycerate-3-phosphate (GP) and triose phosphate (TP) but no other biochemical detail is required. *M1.1, M1.2, M1.3, M1.6, M1.7, M3.5* HSW4, HSW5, HSW6, HSW8
(d)	**(i)** practical investigations into the factors affecting photosynthesis	To include light wavelength, light intensity, light duration, temperature and pH. *M1.1, M1.2, M1.3, M1.6, M1.7, M1.9, M3.5* **PAG4** HSW3, HSW4, HSW5, HSW6
	(ii) practical investigations of the Hill Reaction (light dependent reaction) using DCPIP	*M1.1, M1.2, M1.3, M1.6, M1.7, M3.5* **PAG4** HSW3, HSW4, HSW5, HSW6
(e)	the metabolism of TP and GP to produce carbohydrates, lipids and amino acids	To include reference to the need for appropriate mineral ions (e.g. nitrates and sulfates in relation to amino acids).
(f)	the dependency of respiration in plants and animals upon the products of photosynthesis	
(g)	**(i)** the biological significance of the compensation point for crop production	
	(ii) an interpretation of data and graphs relating to the biological significance of the compensation point for crop production	*M3.4, M3.5*
	(iii) practical investigation of different leaf samples to compare compensation points with hydrogencarbonate indicator solution	*M1.1, M1.2, M1.3, M1.6, M1.7, M3.4, M3.5* HSW3, HSW4, HSW5, HSW6
(h)	**(i)** the importance of microorganisms in maintaining ecosystems, with reference to the nitrogen cycle	To include the recycling of nitrogen by bacteria, with reference to the roles of *Azotobacter*, *Nitrosomonas*, *Nitrobacter* and *Rhizobium* **AND** the use of luminescent microorganisms as biosensors in environmental monitoring. HSW8
	(ii) culturing of *Rhizobium spp. in vitro*	*M1.8* **PAG1** HSW4, HSW5, HSW6
	(iii) investigating the appearance of root nodules in legumes	*M1.8* HSW4, HSW5, HSW6

(i)	the transfer of biomass through a food chain in food production	*M0.4, M1.2, M1.6*
(j)	a consideration of the efficiency of biomass transfers in the food chain with reference to their comparative ability to provide resources in a sustainable fashion	To include fish farming and maize grown as animal feed for beef cattle reared for human consumption.
(k)	the role of ruminants in the human food chain	To include an outline of the structure and composition of the ruminant digestive system as an ecosystem and the role of microorganisms within this ecosystem in the digestion of cellulose, the production of fatty acids and as a source of protein.
(l)	**(i)** farms as ecosystems **(ii)** the potential for conflict between agriculture and conservation	To include a consideration of farms as ecosystems where biotic and abiotic factors impact on productivity **AND** the features of intensive and extensive farming to include the advantages and disadvantages with reference to hedgerow removal, use of chemicals, disposal of farm waste and consideration of government schemes such as the countryside stewardship scheme.
(m)	**(i)** how land management can result in deflected succession	Examples of land management to include forestry and agriculture. HSW9, HSW10, HSW11
	(ii) practical investigations of differences in biodiversity using techniques such as random and systematic sampling.	*M1.5, M1.6, M1.7, M2.3* **PAG3** HSW4, HSW5, HSW6